University of Cape Town Department of Physics

PHY1004W Laboratory Test (Semester 1)

Sets of the apparatus have been set up in PHYLAB1 for you to go and practice today (14h00-17h00) and on 4th and 5th May between 13h00 and 13h55 (no demonstrators available).

Time: 1 hr (data collection) + 2 (report writeup) **Marks: 30**
Group 1 starts at 13h00 and ends at 16h00 in PHYLAB I
Group 2 starts at 14h00 and ends at 17h00 in PHYLAB I

Duration of the lab test:

1. Report to the test venue (PHYLAB1) at least 10 minutes before the start of your test.
2. The total test period is 3 hours. Once the test has started you will have access to the apparatus for the first 60 minutes. You will move to another venue where you will have the remaining time to complete the write-up. During the remaining 120 minutes you will no longer have access to the apparatus.

What you are allowed to take into the lab test venue:

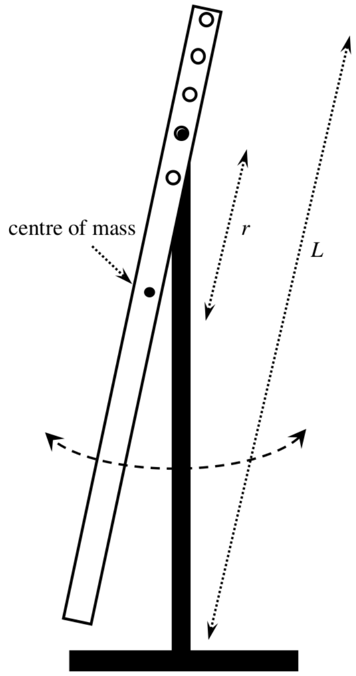
1. Bring your labtop with you.
2. This sheet and python code will be available to you via the Lab Test Lesson on Vula.

Communication during the lab test:

1. You will perform the laboratory test on your own (i.e., no group work).
2. No communication between students will be allowed during the lab test.
3. You may only communicate with the PHYLAB I teaching team.
4. You may not refer to hard copies or electronic copies of "Introduction to Measurement in the Physics Laboratory".
5. You may only use the resources on the Lab Test Vula page and nothing else. Using any other internet resource is a form of cheating.

The Compound Pendulum

A "compound pendulum" has been constructed from a thin wooden metre stick which can be caused to swing on a pin connected to a retort stand which is clamped to the laboratory bench.



Holes have been pre-drilled into the metre stick.

The period T of oscillation of the stick for small amplitude oscillations about an axis perpendicular to the plane of the stick is given by

$$T = 2\pi\sqrt{\frac{I}{mgr}}, \tag{1}$$

where I is the moment of inertia of the stick about the pivot point a distance r from its centre of mass, m is the mass of the stick, and g is the acceleration due to gravity.

Since $I = I_{CM} + mr^2$ where I_{CM} is the moment of inertia of the stick about a parallel axis through the centre of mass,

$$T = 2\pi\sqrt{\frac{I_{CM} + mr^2}{mgr}} \tag{2}$$

which can be transformed to

$$T^2 r = 4\pi^2\left(\frac{I_{CM}}{mg} + \frac{r^2}{g}\right) \tag{3}$$

Since $I_{CM} = \frac{1}{12}mL^2$ for the metre stick, we end up with

$$T^2 r = \left(\frac{4\pi^2}{g}\right) r^2 + \frac{4\pi^2}{12g}L^2 \tag{4}$$

Hang the metre stick from the first hole nearest the end of the metre stick. Cause the stick to swing back and forth with <u>small</u> amplitude. Make careful measurements of T and r (where r is the distance from the centre of mass to the axis of rotation). Repeat for the next 4 holes.

Use the code `LinearLeastSquares2.py` to find a line of best fit through your plotted data, and to complete a least squares analysis of the data in order to obtain best values for the slope and y-intercept (and their associated uncertainties) for a straight line fitted to the data.

Determine values for g from each of the values for the slope and y-intercept.

Values for the standard uncertainties $u(g)$ for the two values of g can be calculated using

$$\frac{u(g)}{g} = \frac{u(\text{slope})}{\text{slope}} \qquad \text{and} \qquad \left(\frac{u(g)}{g}\right)^2 = \left(\frac{u(y\text{-int})}{y\text{-int}}\right)^2 + \left(2\frac{u(L)}{L}\right)^2.$$

Do the two measured values for g agree with each other? Explain carefully.

Type up a full report in which you describe the aim and method succinctly. Present the data and analysis clearly. Make appropriate comments about the quality of your results and how these might be improved.

Upload a copy of your complete report as a **pdf** file using the Lab Test Assignment link **for your group**:
Group 1 must submit a pdf file <u>before</u> 16h00. Name your file PeoplesoftID_Group1.pdf.
Group 2 must submita pdf file <u>before</u> 17h00. Name your file PeoplesoftID_Group2.pdf.

Some hints:

1. Use small swings, not violent rocking. Only use the first 5 holes.
2. Think carefully about what data you need for each hole. How many swings are best?
3. Use your time efficiently. Record your data directly into a spreadsheet (or into `LinearLeastSquares2.py`).
4. Plan your time beforehand. Write your very best report.
5. Take the opportunity to go and practice the lab beforehand and discuss with the demonstrators if you need to.